

From Dystopia to Eutopia: Transforming Urban Environments Through Collaborative Decision-Making in ARSINOE VR

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Figure 1: Left: The dystopian scenario from the flora expert's perspective during scanning. Middle: The spacecraft with the miniature, player choices, and their impact. Right: The eutopian scenario with a synchronization point from the fauna expert's view.

ABSTRACT

The urban heat island effect, as a direct consequence of climate change, poses a significant health risk in urban environments. ARSINOE VR is an innovative social VR experience for environmental education, providing users with an immersive look at potential paths humanity could take regarding environmental protection. Set in a re-imagined city square, the experience immerses two users in alternate realities—a dystopian future and a sustainable alternative—empowering them to explore the effects of climate change and strategize solutions. Participants assume the roles of Flora Expert and Fauna Expert, utilizing interaction mechanics that prioritize collaboration and dialogue. These include environmental scanning based on proprioceptive principles, synchronized embodied actions and miniature manipulation to analyze environmental challenges and mitigation strategies. Through dialogue and collaboration, users engage in meaningful discussions about nature-based solutions, fostering critical thinking and environmental awareness.

Index Terms: social VR, 3D interaction, environmental education.

1 INTRODUCTION

In many large metropolitan cities, the urban heat island effect poses significant health risks. Climate change adaptation is urgently needed, but intense urbanization has reduced city residents' interaction with nature. Traditional environmental education relies on theoretical knowledge and passive learning. Virtual Reality (VR) has been used to present environmental issues [6, 8, 3] and has been found to be effective in promoting pro-environmental behaviors, such as raising awareness about plastic pollution [5, 10]. However, existing VR approaches are predominantly single-user.

ARSINOE VR is a social VR experience that immerses users in potential paths humanity could take for environmental protection. Geared towards two users collaborating in VR, the experience

allows them to explore together two alternative futures—with and without measures for climate change mitigation—through the use of compelling visuals combined with interaction design that prioritizes collaboration and dialogue. ARSINOE VR fosters deeper reflection and meaningful conversations about climate change, aiming to educate participants about nature-based solutions while enhancing environmental awareness, citizenship and empathy.

2 METHODOLOGY AND EXPERIENCE DESIGN

Dialogic and collaborative pedagogy, embedded in the constructivist theory of education, considers collaboration, active participation and dialogue as a process through which participants jointly explore and develop a common understanding of a topic [2, 1]. According to Arizpe Vicencio, collaboration and dialogue are crucial for effective environmental education activities [11], and for promoting awareness of the learners' broader roles as citizens engaged in environmental stewardship.

Our goal when designing ARSINOE VR was to create an immersive VR experience that fosters environmental awareness and understanding of climate change mitigation strategies. With that in mind, we designed appropriate interactions leveraging embodiment, player synchronization and a miniaturized version of the environment. For the implementation, we used the Unity game engine, Meta's SDKs and several other assets and 3D models.

2.1 Story and game flow

Set in a reimagined city square symbolizing the intersection of humanity and the natural world, ARSINOE VR guides two users through a collaborative journey. Wearing a futuristic virtual helmet with a Heads-Up Display (HUD), participants assume the roles of "Flora Expert" or "Fauna Expert," tasked with exploring alternate realities and making impactful decisions.

The journey begins in a spacecraft's training room where users learn the interactions. After learning how to move around and grab objects, the two players are asked to decide which role will be assigned to each of them. To do this, they are presented with two similar crates with different decorations, one flora-related and the other fauna-related. Then, they each open the selected crate to find

the respective helmet for the role. When wearing the helmet, a virtual HUD appears with distinctive visual elements for each role.

After they learn how to scan elements related to their role and how to activate machinery by pressing buttons simultaneously, they can activate a portal leading to a dystopian future city square. Here, they encounter the harsh realities of climate change and observe its detrimental effects on plant and animal life. Users work together to collect general information about the environment by activating a weather station. Then, they alternate in collecting information about the flora and fauna from a set of predefined points-of-interest, sharing and discussing with the other person. After users complete the collaborative exploration of the square, they return to the spacecraft to strategize solutions about the degraded environment.

Back in the spacecraft, ARSINOE (the spacecraft's "AI") asks players to choose between a series of different solutions, such as trees vs artificial shade or cool materials vs green infrastructure, while presented with a miniature of the environment they just visited. Players engage in discussions to prioritize their choices and then they add the selected virtual elements to the miniature, transforming the environment and receiving feedback from ARSINOE.

When the miniature transformation is done, they travel to the square once more to collaboratively collect again information from the same weather station and flora/fauna points-of-interest as before. However, this time they observe and discuss the positive impact of their decisions in this alternate eutopian reality.

2.2 Interaction mechanics

Scanning: We propose a novel interaction method for scanning virtual elements that denote points-of-interest designed to provide relevant information to the players. Based on research for the application of the proprioceptive model of the human body in VR [13], we leverage the ability of humans to locate a specific body part and map it to an action. In our case, we chose to equip the players with a helmet that has the ability to scan specific types of elements depending on the role of the player, either fauna- or flora-related. Each role has unique scanning abilities, encouraging players to collaborate and share knowledge. When a point-of-interest is active, the player with the relevant role can see an "i" icon above the element in question. After they get close to it, players can start the scanning process by bringing their controller close to their ear and pressing the trigger button until the scanning process is completed. Then, relevant textual information appears on the player's HUD. The goal was to stay consistent with the thematic direction of the experience while providing an intuitive and usable interaction mechanic.

HUD: Designing an effective HUD for VR requires balancing usability and user comfort. To prevent fatigue and dizziness commonly linked with HUDs in VR [12], we used minimal and translucent visuals, and implemented smooth placement that follows the user's gaze with a slight latency. This approach avoids inducing motion sickness and ensures visual clarity.

Synchronizing: The multi-user social VR experience Çatalhöyük VR [7] introduced the "high-five" interaction mechanism [4], devised to establish a synchronized flow of collaborative and coordinated action. Inspired by the "high-five" paradigm, ARSINOE VR employs a technology oriented metaphor which requires users to activate machinery together to achieve shared goals. The two users have to collaborate to activate different devices needed to achieve shared goals. This is implemented by both users having to press at the same time a virtual button that is relevant to the role of each one. The button has the distinguishing characteristics of the corresponding role.

Miniature manipulation: Inspired by the world-in-miniature interaction metaphor [9], this mechanic employs a miniature version of the city square, allowing users to visualize and manipulate solutions on a smaller scale, while strategizing together environmental protection approaches in the spacecraft. The miniature en-

hances spatial understanding and acts as a focus for dialogue and joint reflection, as players experiment with different approaches to create a sustainable future.

3 CONCLUSION

ARSINOE VR offers the unique combination of immersion, collaboration and dialogue, which creates a potent learning environment for deeper understanding and critical thinking. It offers an experiential approach, where the pair of users can explore together climate change effects and potential solutions. ARSINOE VR also leverages storytelling to inspire empathy, awareness, and action for climate change mitigation, with collaborative interactions as key features of the experience. Users learn that working together to make small, informed decisions today can shape a sustainable tomorrow.

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